

# Chapter 8 Practice Workbook

*Quadratic Equations — Retake & Rebuild · Neil Velicheti · April 2026*

**How to use this workbook.** Work each problem in order, in pen, on the lines provided. Show every step — no calculator on the work pages. Treat it like the actual test. When you finish a topic, flip to the Solutions section at the back and check yourself. Every problem you missed → write a one-line note in your error notebook (what went wrong, what to remember next time).

**Topics covered.** Same 7 topics that were on the Chapter 8 test you took on April 23. Total: 23 problems. Time budget: about 60 minutes if working fluently.

**If you get stuck.** Try the cheat sheet first. Then check the worked solution. Don't peek before trying.

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## Topic 1 — Graphing Quadratic Equations

Build a table of values, plot, and identify axis of symmetry, vertex, domain, and range. The vertex is on the axis of symmetry  $x = -b/(2a)$ .

### Problem 1.1

Graph  $y = x^2 - 4x + 3$  using a table of values. Identify the axis of symmetry, vertex, domain, and range.

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**Problem 1.2**

Graph  $y = -x^2 + 6x - 8$  using a table of values. Identify the axis of symmetry, vertex, domain, and range.

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**Problem 1.3**

Given  $y = 2(x - 1)^2 + 3$ , identify the axis of symmetry, vertex, direction of opening, and range.

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**Topic 2 — Vertex Form & Transformations**

From  $y = x^2$ , the equation  $y = a(x - h)^2 + k$  transforms it:  $h$  shifts left/right (sign flips),  $k$  shifts up/down,  $|a|$  stretches or compresses, negative  $a$  reflects over the  $x$ -axis.

**Problem 2.1**

Describe how the graph of  $y = (x + 4)^2 - 1$  is transformed from the parent function  $y = x^2$ .

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**Problem 2.2**

Describe how the graph of  $y = -3(x - 2)^2 + 5$  is transformed from  $y = x^2$ .

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**Problem 2.3**

Write the equation of a quadratic that is the parent function  $y = x^2$  reflected over the x-axis, then shifted right 3 and down 4.

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**Problem 2.4**

Write the equation of a quadratic that is the parent function  $y = x^2$  vertically stretched by a factor of 2 and shifted left 5.

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**Topic 3 — Converting Forms + Identifying Zeros**

Convert between vertex form and standard form by expanding the squared binomial, then factor to find the zeros (the x-values where  $y = 0$ ).

**Problem 3.1**

Convert  $y = (x + 2)^2 - 9$  to standard form. Then factor and identify the zeros.

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**Problem 3.2**

Convert  $y = -(x - 3)^2 + 16$  to standard form. Then factor and identify the zeros.

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**Problem 3.3**

Convert  $y = 2(x + 1)^2 - 8$  to factored form and identify the zeros.

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**Topic 4 — Discriminant & Number of Solutions**

The discriminant  $D = b^2 - 4ac$  tells you how many real solutions a quadratic has:  $D > 0 \rightarrow$  two real,  $D = 0 \rightarrow$  one (double) real,  $D < 0 \rightarrow$  no real solutions.

**Problem 4.1**

Find the discriminant of  $y = x^2 + 5x + 7$ . How many real solutions does the equation  $x^2 + 5x + 7 = 0$  have?

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**Problem 4.2**

Find the discriminant of  $y = 4x^2 - 12x + 9$ . How many real solutions does the equation have?

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**Problem 4.3**

Find the discriminant of  $y = -2x^2 + 3x + 4$ . How many real solutions does the equation have?

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**Topic 5 — Solving Quadratics (4 Methods)**

Pick the method that fits the problem. Factoring is fastest when it works. Square roots when there's no  $x$ -term. Completing the square when factoring fails and  $a = 1$ . Quadratic formula always works.

**Problem 5.1**

Solve by factoring:  $x^2 - 5x - 14 = 0$

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**Problem 5.2**

Solve by square roots:  $3(x - 4)^2 = 75$

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**Problem 5.3**

Solve by completing the square:  $x^2 + 6x - 16 = 0$

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**Problem 5.4**

Solve using the quadratic formula:  $2x^2 - 7x + 1 = 0$  (express answers in simplest radical form).

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**Topic 6 — Word Problems (Area & Consecutive Integers)**

Define a variable, translate the words to a quadratic equation, solve. Always check if a solution makes physical sense (lengths can't be negative).

**Problem 6.1**

The area of a rectangle is 60 sq cm. Its length is 4 cm more than its width. Find the dimensions.

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**Problem 6.2**

Two consecutive odd integers have a product of 99. Find them.

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**Problem 6.3**

The base of a triangle is  $x$  and its height is  $x + 5$ . The area is 42 square units. Find  $x$ .

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**Topic 7 — Linear vs. Quadratic Models**

If a data table has constant first differences in  $y$ , the model is linear ( $y = mx + b$ ). If first differences change but second differences are constant, the model is quadratic. For projectile motion:  $h(t) = -16t^2 + v_0 \cdot t + h_0$ .

**Problem 7.1**

Determine if the data is linear or quadratic, write an equation, and predict the  $y$ -value at  $x = 6$ .

$x$ : 1 2 3 4  
 $y$ : 2 5 8 11

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**Problem 7.2**

A ball is thrown upward with initial velocity 48 ft/s from a height of 5 ft. Its height is  $h(t) = -16t^2 + 48t + 5$ . (a) What is the maximum height? (b) When does it hit the ground?

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**Problem 7.3**

Determine if the data is linear or quadratic, write an equation, and predict  $y$  at  $x = 5$ .

$x$ : 0 1 2 3

$y$ : 3 7 15 27

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# Solutions (Worked, Step-by-Step)

Don't read until you've tried the problem yourself. Each step shows the reasoning, not just the math. The boxed line at the end is the final answer.

## Topic 1 — Graphing Quadratic Equations

### Problem 1.1

Graph  $y = x^2 - 4x + 3$  using a table of values. Identify the axis of symmetry, vertex, domain, and range.

Find axis of symmetry:  $x = -b/(2a) = -(-4)/(2 \cdot 1) = 2$ .

Build a table around  $x = 2$ :

$x$ : 0 1 2 3 4

$y$ : 3 0 -1 0 3

Vertex is the lowest point: (2, -1).

Since  $a = 1 > 0$ , opens up. Domain: all real numbers (■). Range:  $y \geq -1$ .

**Answer: Axis  $x = 2$ ; Vertex (2, -1); Domain ■; Range  $y \geq -1$ .**

### Problem 1.2

Graph  $y = -x^2 + 6x - 8$  using a table of values. Identify the axis of symmetry, vertex, domain, and range.

Axis of symmetry:  $x = -b/(2a) = -6/(2 \cdot (-1)) = 3$ .

Table around  $x = 3$ :

$x$ : 1 2 3 4 5

$y$ : -3 0 1 0 -3

Vertex (3, 1).  $a = -1 < 0$ , opens down.

**Answer: Axis  $x = 3$ ; Vertex (3, 1); Domain ■; Range  $y \leq 1$ .**

### Problem 1.3

Given  $y = 2(x - 1)^2 + 3$ , identify the axis of symmetry, vertex, direction of opening, and range.

This is vertex form  $y = a(x - h)^2 + k$  with  $a = 2$ ,  $h = 1$ ,  $k = 3$ .

Vertex is (h, k) = (1, 3).

Axis of symmetry is  $x = h$ , so  $x = 1$ .

$a = 2 > 0$ , so opens up. Range:  $y \geq 3$ .

**Answer: Axis  $x = 1$ ; Vertex (1, 3); Opens up; Range  $y \geq 3$ .**

## Topic 2 — Vertex Form & Transformations

### Problem 2.1

Describe how the graph of  $y = (x + 4)^2 - 1$  is transformed from the parent function  $y = x^2$ .

Rewrite as  $y = (x - (-4))^2 + (-1)$ . So  $h = -4$ ,  $k = -1$ .

$h = -4 \rightarrow$  shift LEFT 4 units.

$k = -1 \rightarrow$  shift DOWN 1 unit.

**Answer: Left 4, down 1.**

**Problem 2.2**

Describe how the graph of  $y = -3(x - 2)^2 + 5$  is transformed from  $y = x^2$ .

Compare to  $y = a(x - h)^2 + k$ :  $a = -3$ ,  $h = 2$ ,  $k = 5$ .

Negative  $a \rightarrow$  reflect over  $x$ -axis.

$|a| = 3 \rightarrow$  vertical stretch by factor of 3.

$h = 2 \rightarrow$  right 2.  $k = 5 \rightarrow$  up 5.

**Answer: Reflect over  $x$ -axis, vertical stretch by 3, right 2, up 5.**

**Problem 2.3**

Write the equation of a quadratic that is the parent function  $y = x^2$  reflected over the  $x$ -axis, then shifted right 3 and down 4.

Reflection over  $x$ -axis:  $a = -1$ .

Right 3  $\rightarrow h = 3$ . Down 4  $\rightarrow k = -4$ .

**Answer:  $y = -(x - 3)^2 - 4$**

**Problem 2.4**

Write the equation of a quadratic that is the parent function  $y = x^2$  vertically stretched by a factor of 2 and shifted left 5.

Stretch factor 2:  $a = 2$ .

Left 5  $\rightarrow h = -5$ . No vertical shift  $\rightarrow k = 0$ .

**Answer:  $y = 2(x + 5)^2$**

## Topic 3 — Converting Forms + Identifying Zeros

**Problem 3.1**

Convert  $y = (x + 2)^2 - 9$  to standard form. Then factor and identify the zeros.

Expand the square:  $(x + 2)^2 = x^2 + 4x + 4$ .

$y = x^2 + 4x + 4 - 9 = x^2 + 4x - 5$ .

Factor: need two numbers with product  $-5$ , sum 4  $\rightarrow 5$  and  $-1$ .

$y = (x + 5)(x - 1)$ .

Zeros:  $x + 5 = 0 \rightarrow x = -5$ ;  $x - 1 = 0 \rightarrow x = 1$ .

**Answer: Standard form:  $y = x^2 + 4x - 5$ . Zeros:  $x = -5, 1$ .**

**Problem 3.2**

Convert  $y = -(x - 3)^2 + 16$  to standard form. Then factor and identify the zeros.

$(x - 3)^2 = x^2 - 6x + 9$ .

$y = -(x^2 - 6x + 9) + 16 = -x^2 + 6x - 9 + 16 = -x^2 + 6x + 7$ .

Factor out  $-1$ :  $y = -(x^2 - 6x - 7)$ .

Factor  $x^2 - 6x - 7$ : product  $-7$ , sum  $-6 \rightarrow -7$  and  $1$ . So  $(x - 7)(x + 1)$ .

$y = -(x - 7)(x + 1)$ . Zeros where each factor = 0:  $x = 7$ ,  $x = -1$ .

**Answer: Standard form:  $y = -x^2 + 6x + 7$ . Zeros:  $x = -1, 7$ .**

**Problem 3.3**

Convert  $y = 2(x + 1)^2 - 8$  to factored form and identify the zeros.

Expand:  $2(x + 1)^2 - 8 = 2(x^2 + 2x + 1) - 8 = 2x^2 + 4x + 2 - 8 = 2x^2 + 4x - 6$ .

Factor out GCF 2:  $y = 2(x^2 + 2x - 3)$ .

Factor  $x^2 + 2x - 3$ : product  $-3$ , sum  $2 \rightarrow 3$  and  $-1$ . So  $(x + 3)(x - 1)$ .

$y = 2(x + 3)(x - 1)$ . Zeros:  $x = -3, 1$ .

**Answer: Factored:  $y = 2(x + 3)(x - 1)$ . Zeros:  $x = -3, 1$ .**

**Topic 4 — Discriminant & Number of Solutions****Problem 4.1**

Find the discriminant of  $y = x^2 + 5x + 7$ . How many real solutions does the equation  $x^2 + 5x + 7 = 0$  have?

$a = 1, b = 5, c = 7$ .

$D = b^2 - 4ac = 5^2 - 4(1)(7) = 25 - 28 = -3$ .

**Answer:  $D = -3$  (negative)  $\rightarrow$  0 real solutions.**

**Problem 4.2**

Find the discriminant of  $y = 4x^2 - 12x + 9$ . How many real solutions does the equation have?

$a = 4, b = -12, c = 9$ .

$D = (-12)^2 - 4(4)(9) = 144 - 144 = 0$ .

**Answer:  $D = 0 \rightarrow$  1 real (double) solution.**

**Problem 4.3**

Find the discriminant of  $y = -2x^2 + 3x + 4$ . How many real solutions does the equation have?

$a = -2, b = 3, c = 4$ .

$D = 3^2 - 4(-2)(4) = 9 + 32 = 41$ .

**Answer:  $D = 41$  (positive)  $\rightarrow$  2 real solutions.**

**Topic 5 — Solving Quadratics (4 Methods)****Problem 5.1**

Solve by factoring:  $x^2 - 5x - 14 = 0$

Need two numbers with product  $-14$ , sum  $-5 \rightarrow -7$  and  $2$ .

$(x - 7)(x + 2) = 0$ .

Zero-product:  $x - 7 = 0$  OR  $x + 2 = 0$ .

**Answer:  $x = 7$  or  $x = -2$ .**

**Problem 5.2**

Solve by square roots:  $3(x - 4)^2 = 75$

Divide both sides by 3:  $(x - 4)^2 = 25$ .

Square root both sides:  $x - 4 = \pm 5$ .

$x = 4 + 5 = 9$  or  $x = 4 - 5 = -1$ .

**Answer:  $x = 9$  or  $x = -1$ .**

**Problem 5.3**

Solve by completing the square:  $x^2 + 6x - 16 = 0$

Move c:  $x^2 + 6x = 16$ .

Half of b is 3. Square it: 9. Add 9 to BOTH sides.

$x^2 + 6x + 9 = 16 + 9 \rightarrow (x + 3)^2 = 25$ .

Square root:  $x + 3 = \pm 5$ .

$x = -3 + 5 = 2$  or  $x = -3 - 5 = -8$ .

**Answer:  $x = 2$  or  $x = -8$ .**

**Problem 5.4**

Solve using the quadratic formula:  $2x^2 - 7x + 1 = 0$  (express answers in simplest radical form).

$a = 2$ ,  $b = -7$ ,  $c = 1$ .

Discriminant:  $(-7)^2 - 4(2)(1) = 49 - 8 = 41$ .

$x = (-b \pm \sqrt{D}) / (2a) = (7 \pm \sqrt{41}) / 4$

**Answer:  $x = (7 + \sqrt{41})/4$  or  $x = (7 - \sqrt{41})/4$ .**

## Topic 6 — Word Problems (Area & Consecutive Integers)

**Problem 6.1**

The area of a rectangle is 60 sq cm. Its length is 4 cm more than its width. Find the dimensions.

Let  $w$  = width. Then length =  $w + 4$ .

Area:  $w(w + 4) = 60 \rightarrow w^2 + 4w - 60 = 0$ .

Factor: product  $-60$ , sum  $4 \rightarrow 10$  and  $-6$ .  $(w + 10)(w - 6) = 0$ .

$w = -10$  (reject — negative) or  $w = 6$ .

Width 6, length  $6 + 4 = 10$ .

**Answer: Dimensions: 6 cm by 10 cm.**

**Problem 6.2**

Two consecutive odd integers have a product of 99. Find them.

Let  $n$  = first odd integer. Then  $n + 2$  is the next odd integer.

$n(n + 2) = 99 \rightarrow n^2 + 2n - 99 = 0$ .

Factor: product  $-99$ , sum  $2 \rightarrow 11$  and  $-9$ .  $(n + 11)(n - 9) = 0$ .

$n = -11 \rightarrow$  integers  $-11$  and  $-9$ .  $n = 9 \rightarrow$  integers  $9$  and  $11$ .

**Answer: Either  $\{9, 11\}$  or  $\{-11, -9\}$ .**

**Problem 6.3**

The base of a triangle is  $x$  and its height is  $x + 5$ . The area is 42 square units. Find  $x$ .

Triangle area =  $(1/2) \cdot \text{base} \cdot \text{height}$ .

$$(1/2) \cdot x \cdot (x + 5) = 42 \rightarrow x(x + 5) = 84 \rightarrow x^2 + 5x - 84 = 0.$$

Factor: product  $-84$ , sum  $5 \rightarrow 12$  and  $-7$ .  $(x + 12)(x - 7) = 0$ .

$$x = -12 \text{ (reject) or } x = 7.$$

**Answer:  $x = 7$ .**

## Topic 7 — Linear vs. Quadratic Models

**Problem 7.1**

Determine if the data is linear or quadratic, write an equation, and predict the  $y$ -value at  $x = 6$ .

$x$ : 1 2 3 4

$y$ : 2 5 8 11

First differences in  $y$ :  $5-2=3$ ,  $8-5=3$ ,  $11-8=3$ . Constant  $\rightarrow$  LINEAR.

Slope  $m = 3$ . Using point  $(1, 2)$ :  $y - 2 = 3(x - 1) \rightarrow y = 3x - 1$ .

At  $x = 6$ :  $y = 3(6) - 1 = 17$ .

**Answer: Linear;  $y = 3x - 1$ ; at  $x = 6$ ,  $y = 17$ .**

**Problem 7.2**

A ball is thrown upward with initial velocity 48 ft/s from a height of 5 ft. Its height is  $h(t) = -16t^2 + 48t + 5$ . (a) What is the maximum height? (b) When does it hit the ground?

(a) Max at  $t = -b/(2a) = -48/(2 \cdot (-16)) = 1.5$  sec.

$$h(1.5) = -16(1.5)^2 + 48(1.5) + 5 = -16(2.25) + 72 + 5 = -36 + 77 = 41 \text{ ft.}$$

(b) Hits ground when  $h(t) = 0$ :  $-16t^2 + 48t + 5 = 0$ .

Use quadratic formula with  $a=-16$ ,  $b=48$ ,  $c=5$ .

$$t = \frac{-48 \pm \sqrt{(48)^2 - 4(-16)(5)}}{2 \cdot (-16)}$$

$$= \frac{-48 \pm \sqrt{(2304 + 320)}}{-32}$$

$$= \frac{-48 \pm \sqrt{2624}}{-32} \approx \frac{-48 \pm 51.22}{-32}$$

Take the positive root:  $t \approx (-48 - 51.22)/(-32) \approx 3.10$  sec.

**Answer: (a) Max height  $\approx 41$  ft. (b) Ground at  $t \approx 3.10$  sec.**

**Problem 7.3**

Determine if the data is linear or quadratic, write an equation, and predict  $y$  at  $x = 5$ .

$x$ : 0 1 2 3

$y$ : 3 7 15 27

First differences:  $7-3=4$ ,  $15-7=8$ ,  $27-15=12$ . Not constant.

Second differences:  $8-4=4$ ,  $12-8=4$ . Constant  $\rightarrow$  QUADRATIC.

For  $y = ax^2 + bx + c$  with second-diff =  $2a \rightarrow a = 2$ .

Use 3 points to solve for  $b$  and  $c$ , e.g. with  $c = y(0) = 3$ , plug  $(1, 7)$ :  $2 + b + 3 = 7 \rightarrow b = 2$ .

Equation:  $y = 2x^2 + 2x + 3$ .

At  $x = 5$ :  $y = 2(25) + 2(5) + 3 = 50 + 10 + 3 = 63$ .

**Answer: Quadratic;  $y = 2x^2 + 2x + 3$ ; at  $x = 5$ ,  $y = 63$ .**

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